

Lab 1: Transformations and Camera Projections (2h)

CGR 2026: Tuesday 1510

Instructions

This laboratory consists of five questions.

- Please ensure that you have downloaded this problem sheet during your lab slot.
- You are encouraged to read through the questions beforehand, but please leave the active solving for the session.
- Discussion is encouraged, but please write your own answers. Keep your answers concise.
- Submit handwritten answers to the tutor before the end of the lab. Show your working.
- Please put your name and student number on your submission.
- You may use lecture material or web resources to look up information.
- You may **not use generative AI tools** to obtain, generate, or check answers.

Question 1 — Translation and Homogeneous Coordinates

(a) Homogeneous coordinates

Explain why a 3D point is represented as

$$\begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

rather than simply

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

when using homogeneous transformation matrices.

What is the significance of representing a direction vector as

$$\begin{bmatrix} x \\ y \\ z \\ 0 \end{bmatrix} ?$$

(b) Translation matrix

Derive the homogeneous-coordinate translation matrix

$$T(t_x, t_y, t_z)$$

which satisfies

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = T \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

where

$$x' = x + t_x, \quad y' = y + t_y, \quad z' = z + t_z.$$

Show your reasoning.

(c) Numerical calculation

Consider

$$\mathbf{p} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 1 \end{bmatrix}$$

and a translation $(t_x, t_y, t_z) = (4, -2, 5)$. Calculate $T\mathbf{p}$ by explicitly performing the matrix multiplication.

(d) Transformation order

Consider a transformation that first rotates a point by 90° around the z axis and then translates it by $(2, 0, 0)$.

Write the complete transformation as a single matrix.

Explain why the order of the matrices is important.

Question 2 — Rotation and Scale

Consider a 3D point represented using homogeneous coordinates:

$$\mathbf{p} = \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}.$$

(a) Rotation

A rotation around the z axis through angle θ is defined by

$$x' = x \cos \theta - y \sin \theta,$$

$$y' = x \sin \theta + y \cos \theta,$$

$$z' = z.$$

Derive the 4×4 homogeneous-coordinate matrix $R_z(\theta)$ such that

$$\mathbf{p}' = R_z(\theta)\mathbf{p}.$$

Show how the entries of the matrix follow from the equations above.

(b) Scaling

Derive the homogeneous-coordinate matrix S which performs

$$(x, y, z) \longrightarrow (s_x x, s_y y, s_z z).$$

State the special case for uniform scaling.

(c) Combined transformation

A point is first scaled and then rotated around the z axis.

Write down a single matrix M such that

$$\mathbf{p}' = M\mathbf{p}.$$

Clearly state which transformation is applied first and explain why the matrix multiplication appears in the order you give.

(d) Numerical calculation

Let

$$\mathbf{p} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}.$$

First scale the point uniformly by 2, then rotate it by 90° around the z axis. Calculate the final point by hand.

Question 3 — Orthographic Projection

Consider a camera at the origin looking along the negative z axis.

For a simple orthographic projection, let

$$x' = sx, \quad y' = sy.$$

The depth z does not affect the projected size.

(a) Derivation

Construct a homogeneous-coordinate matrix P_{ortho} which produces

$$x' = sx, \quad y' = sy, \quad z' = z.$$

Explain which part of the matrix causes the z coordinate to have no effect on the projected x and y coordinates.

(b) Projection of points

Let

$$s = 2.$$

Calculate the projected coordinates of

$$p_1 = (1, 2, -2)$$

and

$$p_2 = (1, 2, -10).$$

(c) Interpretation

The two points have identical x and y coordinates but different depths.

Explain why they have the same projected position under orthographic projection.

(d) Conceptual comparison

Give one advantage of orthographic projection and one situation in which orthographic projection is useful.

Question 4 — Perspective Projection

Consider a pinhole camera at the origin looking along the negative z axis.

A point in camera coordinates is

$$p = (x, y, z), \quad z < 0.$$

Let the focal length of the camera be f .

(a) Derivation

Using similar triangles, derive the perspective projection equation

$$x' = f \frac{x}{-z}.$$

Your answer should include a diagram or a clearly stated similar-triangle relationship.

(b) Vertical coordinate

Using the same reasoning, derive

$$y' = f \frac{y}{-z}.$$

(c) Numerical calculation

Let

$$f = 2.$$

Calculate the projected coordinates of

$$p = (2, 1, -4).$$

(d) Depth and apparent size

Consider

$$p_1 = (2, 0, -2), \quad p_2 = (2, 0, -8).$$

Calculate x'_1 and x'_2 when $f = 1$.

Explain why the point farther from the camera has a smaller projected coordinate.

(e) Orthographic versus perspective

State the mathematical difference between orthographic and perspective projection.

Identify the term in the perspective equation that causes distant objects to appear smaller.

Question 5 — Constructing a Camera Ray from a Pixel

Consider a virtual pinhole camera based on the camera model used in *Physically Based Rendering (PBRT)*.

The camera has:

- world-space position $\mathbf{C}$;
- camera-to-world rotation matrix R ;
- vertical field of view θ ;
- image resolution $W \times H$.

The camera coordinate system is defined so that the camera looks along the negative z axis. The pinhole is at the camera-space origin, and the image plane is located at $z = -1$.

Pixel $(0, 0)$ is the top-left pixel. Pixel coordinates increase to the right and downwards.

You will derive the ray corresponding to a given pixel.

(a) Pixel centre

Let (i, j) be a pixel.

The ray should pass through the centre of the pixel rather than its top-left corner.

Define normalized image coordinates $(u, v) \in [0, 1]^2$ for the centre of pixel (i, j) .

Give expressions for u and v in terms of i, j, W , and H .

(b) Image plane

The vertical field of view is θ and the image plane is one unit in front of the camera.

- Derive the height h of the image plane.
- Derive its width w , taking the image aspect ratio into account.

(c) Pixel to camera-space point

The image plane is centred on the camera's optical axis.

Using (u, v) and the dimensions w and h , derive the camera-space coordinates of the point corresponding to the centre of pixel (i, j) .

Call this point $\mathbf{p}_c$.

Write your answer in the form

$$\mathbf{p}_c = \begin{bmatrix} x_c \\ y_c \\ -1 \end{bmatrix}.$$

Take care to account for the fact that pixel v increases downwards, while camera-space y increases upwards.

(d) Camera-space ray

The ray originates at the pinhole

$$\mathbf{o}_c = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Using $\mathbf{p}_c$, determine the ray direction $\mathbf{d}_c$.

Then normalize the direction to obtain the unit vector $\hat{\mathbf{d}}_c$.

Write the camera-space ray in the form

$$\mathbf{r}_c(t) = \mathbf{o}_c + t\hat{\mathbf{d}}_c, \quad t \geq 0.$$

(e) Camera space to world space

The camera-to-world transformation of a point is

$$\mathbf{p}_w = R\mathbf{p}_c + \mathbf{C}.$$

Determine how a camera-space direction vector is transformed into world space. Hence write expressions for the world-space ray origin $\mathbf{o}_w$ and unit direction $\hat{\mathbf{d}}_w$. Give the final ray in world space:

$$\mathbf{r}_w(t) = \mathbf{o}_w + t\hat{\mathbf{d}}_w.$$

(f) Numerical example

Consider a camera with

$$\mathbf{C} = \begin{bmatrix} 1 \\ 4 \\ 5 \end{bmatrix}, \quad R = I,$$

$$\theta = 90^\circ, \quad W = 400, \quad H = 400.$$

For the pixel

$$(i, j) = (199, 199),$$

calculate the corresponding world-space ray.

Give your final answer as

$$\mathbf{r}_w(t) = \mathbf{o}_w + t\hat{\mathbf{d}}_w.$$

Show the intermediate values from parts (a)–(e).